YEAR 10 MATHS · CHAPTER 6 · MIND MAP

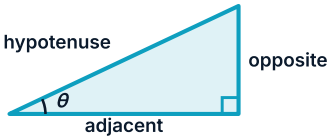
Trigonometry

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1 Trigonometric ratios

SOH CAH TOA.

Hypotenuse 12, angle 50°: opposite $= 12\sin 50° \approx 9.19$



2 Finding unknown angles

Use the inverse buttons: $\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$.

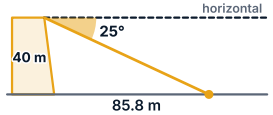
$\cos\theta = \frac{5}{8} \Rightarrow \theta = \cos^{-1}(0.625) \approx 51.3°$

3 Applications in 2D

Elevation and depression are measured from the horizontal.

40 m cliff, angle of depression 25° to a boat

Distance $= \frac{40}{\tan 25°} \approx 85.8$ m

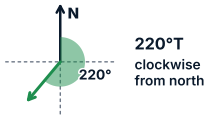


4 Directions and bearings

True bearings: clockwise from north, three digits. Back bearing: add or subtract 180°.

N30°E = 030°T S40°W = 220°T

Back bearing of 060°T is **240°T**

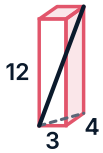


5 Applications in 3D

Find the right triangle in the plane you need, often using Pythagoras first.

Box 3 by 4 by 12: base diagonal 5

Angle with the base: $\tan\theta = \frac{12}{5} \Rightarrow \theta \approx 67.4°$



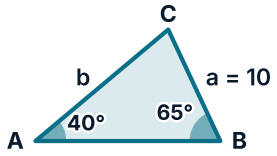
Floor diagonal 5

Space diagonal 13

6 The sine rule

$\frac{a}{\sin A} = \frac{b}{\sin B}$: use it with a side and its opposite angle.

$A = 40°$, $B = 65°$, $a = 10$: $b = \frac{10\sin 65°}{\sin 40°} \approx 14.1$

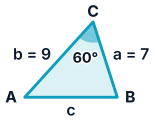


Watch out: finding an angle can give two answers (the ambiguous case).

7 The cosine rule

$c^2 = a^2 + b^2 - 2ab\cos C$: use it with two sides and the angle between, or three sides.

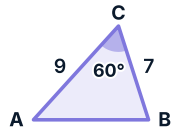
$a = 7$, $b = 9$, $C = 60°$: $c^2 = 49 + 81 - 63 = 67$, so $c \approx 8.19$



8 Area of a triangle

$A = \frac{1}{2}ab\sin C$, with C between sides a and b.

$a = 7$, $b = 9$, $C = 60°$: $A = \frac{1}{2} \times 63 \times \sin 60° \approx 27.3$

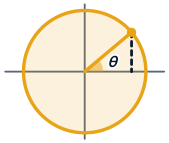


9 The unit circle

A point on the unit circle is $(\cos\theta, \sin\theta)$.

$\sin(180° - \theta) = \sin\theta$ $\cos(180° - \theta) = -\cos\theta$

$\sin 150° = \frac{1}{2}$ $\cos 120° = -\frac{1}{2}$

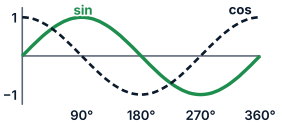


(cos θ, sin θ)
radius 1

10 Graphs of trig functions

$y = \sin x$ and $y = \cos x$: period 360°, between −1 and 1.

$y = \tan x$: period 180°, asymptotes at 90° and 270°



11 Exact values and trig equations

$\sin 30° = \frac{1}{2}$, $\cos 45° = \frac{\sqrt{2}}{2}$, $\tan 60° = \sqrt{3}$

Solve $\sin x = \frac{1}{2}$ for $0° \le x \le 360°$: $x = 30°$ or $150°$

Watch out: check every quadrant where the ratio has the right sign.